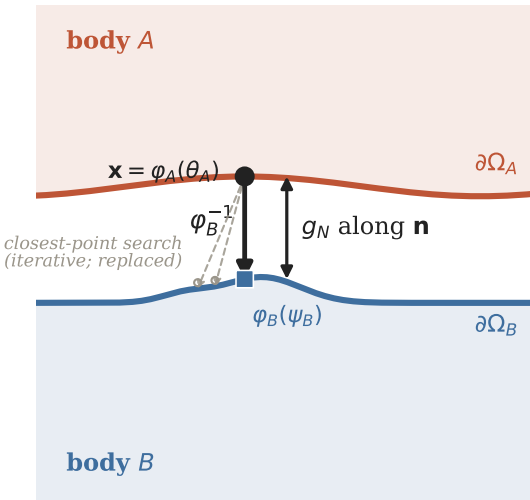
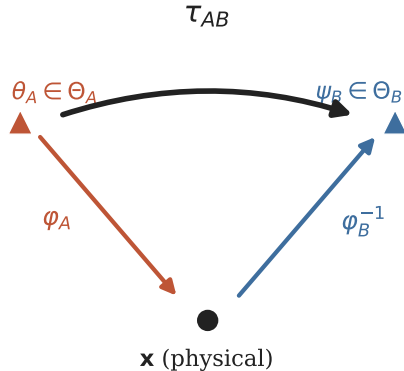


(a) physical coordinates



(b) chart coordinates: the transition map



$$\tau_{AB} = \varphi_B^{-1} \circ \varphi_A = \pi_B \circ \varphi_A$$

one chart evaluation + one inversion: no search